# QCPL-341H

2.5 Amp Output Current IGBT Gate Drive Optocoupler

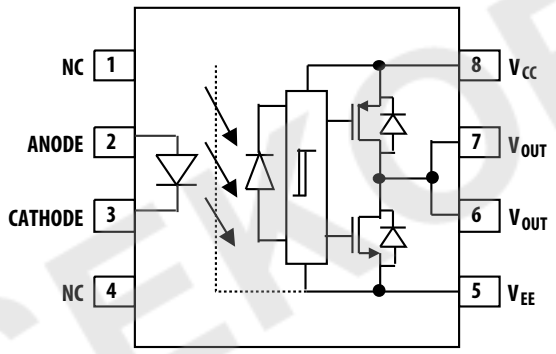
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## Data Sheet

### Description

The QCPL-341H contains an AlGaAs LED, which is optically coupled to an integrated circuit with a power output stage. This optocoupler is ideally suited for driving power IGBTs and MOSFETs used in motor control inverter applications. The high operating voltage range of the output stage provides the drive voltages required by gate controlled devices. The voltage and high peak output current supplied by this optocoupler make it ideally suited for direct driving IGBT with ratings up to 1200V/100A. For IGBTs with higher ratings, this optocoupler can be used to drive a discrete power stage which drives the IGBT gate. The QCPL-341H has the highest insulation voltage of $V_{IORM}$= 630Vpeak in the IEC/ EN/DIN EN 60747-5-5.

### Functional Diagram




### Truth Table

| LED | $V_{CC} - V_{EE}$ "POSITIVE GOING" (i.e., TURN-ON) | $V_{CC} - V_{EE}$ "NEGATIVE GOING" (i.e., TURN-OFF) | $V_O$ |
|---|---|---|---|
| OFF | 0 - 30 V | 0 – 30 V | LOW |
| ON | 0 – 12.1 V | 0 – 11.1 V | LOW |
| ON | 12.1 - 13.5 V | 11.1 – 12.5 V | TRANSITION |
| ON | 13.5 – 30 V | 12.5 – 30 V | HIGH |

Note: Design Note: A 1 µF bypass capacitor must be connected between pins $V_{CC}$ and $V_{EE}$.

### Features

- 2.5 A maximum peak output current
- 2.0 A minimum peak output current
- Rail-to-rail output voltage
- 300 ns maximum propagation delay
- 200 ns maximum propagation delay difference
- LED current input with hysteresis
- 25 kV/µs minimum Common Mode Rejection (CMR) at $V_{CM}$ = 1500 V
- $I_{CC}$ = 5.0 mA maximum supply current
- Under Voltage Lock-Out protection (UVLO) with hysteresis
- Wide operating $V_{CC}$ Range: 15 to 30 V
- Industrial temperature range: -40 °C to 105 °C
- Safety Approval
  - UL Recognized 3750 $V_{RMS}$ for 1min.
  - CSA
  - IEC/EN/DIN EN 60747-5-5 $V_{IORM}$ = 630 Vpeak

### Applications

- IGBT/MOSFET gate drive
- AC and Brushless DC motor drives
- Renewable energy inverters
- Industrial inverters
- Switching power supplies

***CAUTION:*** *It is advised that normal static precautions be taken in handling and assembly of this component to prevent damage and/or degradation which may be induced by ESD.*

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## Ordering Information

QCPL-341H is UL Recognized with 3750 $V_{RMS}$ for 1 minute per UL1577.

| Part number | Option RoHS Compliant | Package | Surface Mount Gull Wing | Tape & Reel | IEC/EN/DIN EN 60747-5-5 | Quantity |
|---|---|---|---|---|---|---|
| QCPL-341H | -000E | 300mil DIP-8 | | | | 50 per tube |
| | -300E | | X | | | 50 per tube |
| | -500E | | X | X | | 1000 per reel |
| | -060E | | | | X | 50 per tube |
| | -360E | | X | | X | 50 per tube |
| | -560E | | X | X | X | 1000 per reel |

To order, choose a part number from the part number column and combine with the desired option from the option column to form an order entry.

Example 1:

QCPL-341H-560E to order product of 300 mil DIP Gull Wing Surface Mount package in Tape and Reel packaging with IEC/EN/DIN EN 60747-5-5 Safety Approval and RoHS compliant.

Example 2:

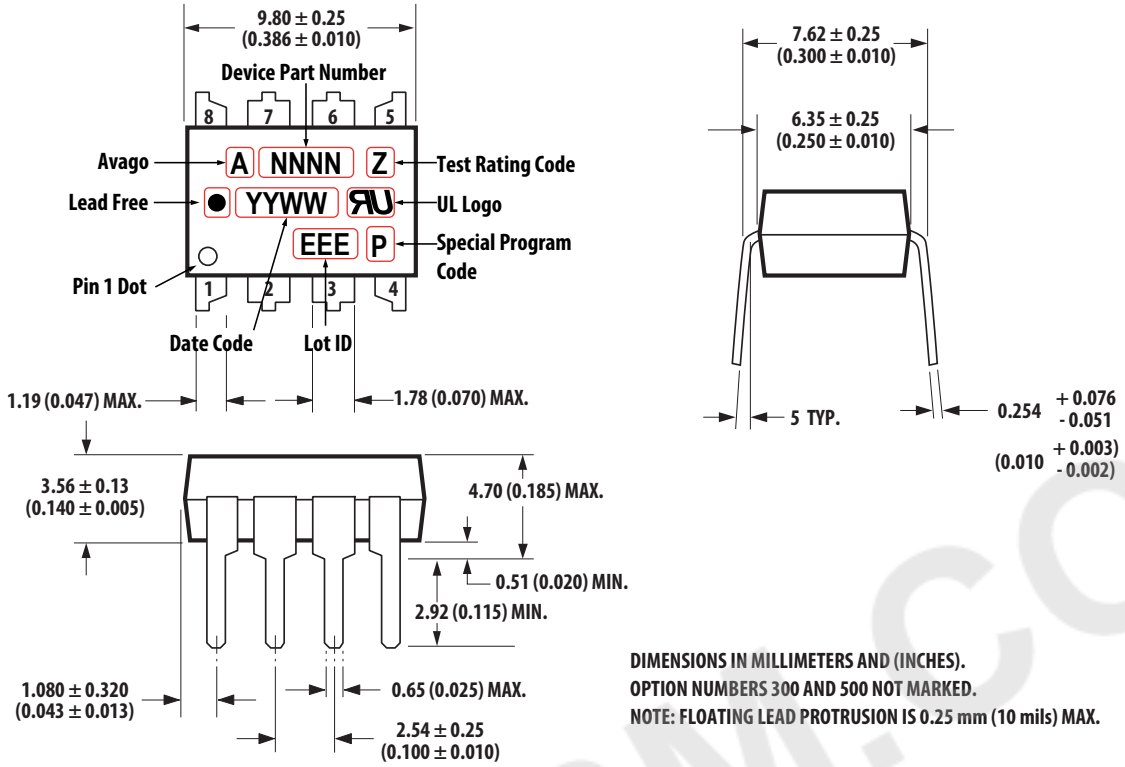
QCPL-341H-000E to order product of 300 mil DIP package in Tube packaging and RoHS compliant.

Option datasheets are available. Contact your Avago sales representative or authorized distributor for information.

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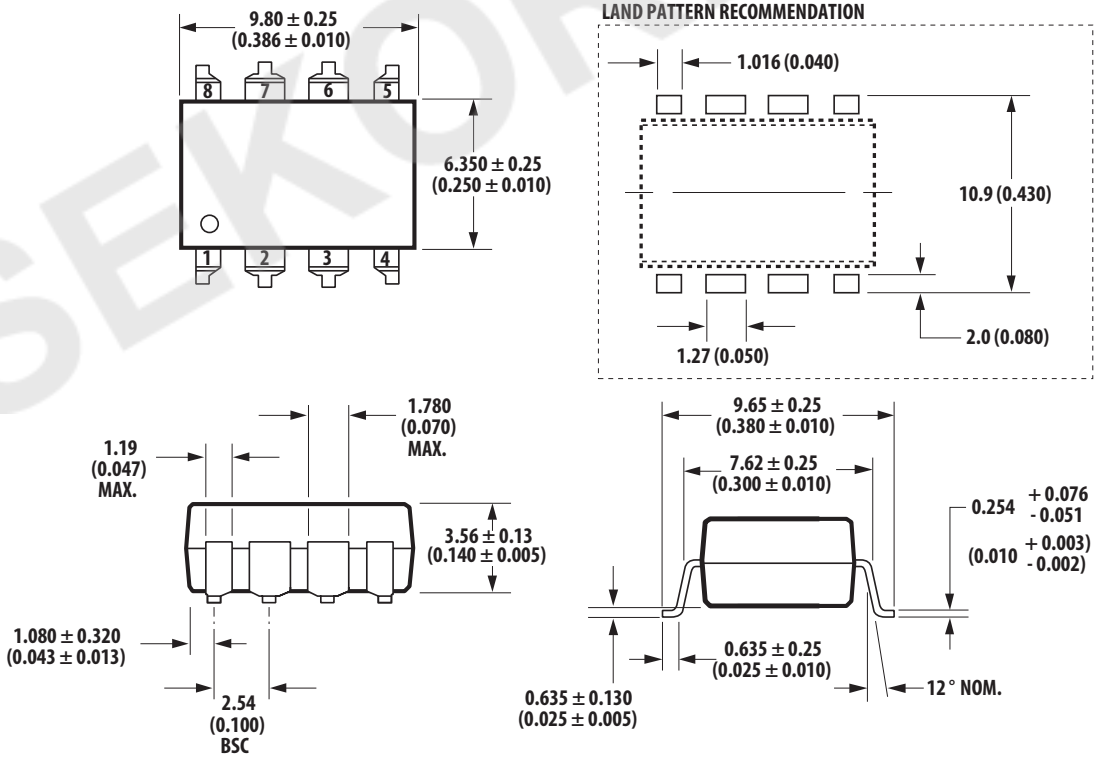
## Package Outline Drawings

### QCPL-341H Outline Drawing (Standard 300mil DIP Package)




DIMENSIONS IN MILLIMETERS AND (INCHES).
OPTION NUMBERS 300 AND 500 NOT MARKED.
NOTE: FLOATING LEAD PROTRUSION IS 0.25 mm (10 mils) MAX.

### QCPL-341H Gull Wing Surface Mount Option 300 Outline Drawing




DIMENSIONS IN MILLIMETERS (INCHES).
LEAD COPLANARITY = 0.10 mm (0.004 INCHES).
NOTE: FLOATING LEAD PROTRUSION IS 0.5 mm (20 mils) MAX.

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## Recommended Pb-Free IR Profile

Recommended reflow condition as per JEDEC Standard, J-STD-020 (latest revision). Non- Halide Flux should be used.

## Regulatory Information

The QCPL-341H is approved by the following organizations:

**UL** Recognized under UL 1577, component recognition program, category, File E55361 up to $V_{ISO}$ = 3750 $V_{RMS}$.

**CSA** CSA Component Acceptance Notice #5, File CA 88324

**IEC/EN/DIN EN 60747-5-5 (Option 060 Only)** Maximum Working Insulation Voltage $V_{IORM}$ = 630Vpeak

**Table 1. IEC/EN/DIN EN 60747-5-5 Insulation Characteristics* (Option 060)**

| Description | Symbol | QCPL-341H Option 060 | Unit |
|---|---|---|---|
| Installation classification per DIN VDE 0110/39, Table 1 | | | |
| for rated mains voltage ≤ 150 $V_{rms}$ | | I – IV | |
| for rated mains voltage ≤ 300 $V_{rms}$ | | I – IV | |
| for rated mains voltage ≤ 450 $V_{rms}$ | | I – III | |
| Climatic Classification | | 40/105/21 | |
| Pollution Degree (DIN VDE 0110/39) | | 2 | |
| Maximum Working Insulation Voltage | $V_{IORM}$ | 630 | $V_{peak}$ |
| Input to Output Test Voltage, Method b* $V_{IORM}$ x 1.875=$V_{PR}$, 100% Production Test with $t_m$=1 sec, Partial discharge < 5 pC | $V_{PR}$ | 1181 | $V_{peak}$ |
| Input to Output Test Voltage, Method a* $V_{IORM}$ x 1.6=$V_{PR}$, Type and Sample Test, $t_m$=10 sec, Partial discharge < 5 pC | $V_{PR}$ | 1008 | $V_{peak}$ |
| Highest Allowable Overvoltage* (Transient Overvoltage tini = 60 sec) | $V_{IOTM}$ | 6000 | $V_{peak}$ |
| Safety-limiting values – maximum values allowed in the event of a failure | | | |
| Case Temperature | $T_S$ | 175 | °C |
| Input Current | $I_{S, INPUT}$ | 230 | mA |
| Output Power | $P_{S, OUTPUT}$ | 600 | mW |
| Insulation Resistance at $T_S$, $V_{IO}$ = 500 V | $R_S$ | $>10^9$ | Ω |

* Refer to the optocoupler section of the Isolation and Control Components Designer's Catalog, under Product Safety Regulations section, (IEC/EN/DIN EN 60747-5-5) for a detailed description of Method a and Method b partial discharge test profiles.

Note: These optocouplers are suitable for "safe electrical isolation" only within the safety limit data. Maintenance of the safety data shall be ensured by means of protective circuits. Surface mount classification is Class A in accordance with CECC 00802.

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**Table 2. Insulation and Safety Related Specifications**

| Parameter | Symbol | QCPL-341H | Units | Conditions |
|---|---|---|---|---|
| Minimum External Air Gap (Clearance) | L(101) | 7.1 | mm | Measured from input terminals to output terminals, shortest distance through air. |
| Minimum External Tracking (Creepage) | L(102) | 7.4 | mm | Measured from input terminals to output terminals, shortest distance path along body. |
| Minimum Internal Plastic Gap (Internal Clearance) | | 0.08 | mm | Through insulation distance conductor to conductor, usually the straight line distance thickness between the emitter and detector. |
| Tracking Resistance (Comparative Tracking Index) | CTI | > 175 | V | DIN IEC 112/VDE 0303 Part 1 |
| Isolation Group | | IIIa | | Material Group (DIN VDE 0110, 1/89, Table 1) |

Notes:
1. All Avago data sheets report the creepage and clearance inherent to the optocoupler component itself. These dimensions are needed as a starting point for the equipment designer when determining the circuit insulation requirements. However, once mounted on a printed circuit board, minimum creepage and clearance requirements must be met as specified for individual equipment standards. For creepage, the shortest distance path along the surface of a printed circuit board between the solder fillets of the input and output leads must be considered (the recommended Land Pattern does not necessarily meet the minimum creepage of the device). There are recommended techniques such as grooves and ribs which may be used on a printed circuit board to achieve desired creepage and clearances. Creepage and clearance distances will also change depending on factors such as pollution degree and insulation level.

**Table 3. Absolute Maximum Ratings**

| Parameter | Symbol | Min. | Max. | Units | Note |
|---|---|---|---|---|---|
| Storage Temperature | $T_S$ | -55 | 125 | °C | |
| Operating Temperature | $T_A$ | -40 | 105 | °C | |
| Output IC Junction Temperature | $T_J$ | | 125 | °C | |
| Average Input Current | $I_{F(AVG)}$ | | 25 | mA | 1 |
| Peak Transient Input Current (<1 µs pulse width, 300pps) | $I_{F(TRAN)}$ | | 1 | A | |
| Reverse Input Voltage | $V_R$ | | 5 | V | |
| "High" Peak Output Current | $I_{OH(PEAK)}$ | | 2.5 | A | 2 |
| "Low" Peak Output Current | $I_{OL(PEAK)}$ | | 2.5 | A | 2 |
| Total Output Supply Voltage | ($V_{CC}$ - $V_{EE}$) | 0 | 35 | V | |
| Output Voltage | $V_{O(PEAK)}$ | -0.5 | $V_{CC}$ | V | |
| Output IC Power Dissipation | $P_O$ | | 250 | mW | 3 |
| Total Power Dissipation | $P_T$ | | 295 | mW | 4 |
| Lead Solder Temperature | 260°C for 10 sec., 1.6 mm below seating plane | | | | |

**Table 4. Recommended Operating Conditions**

| Parameter | Symbol | Min | Max. | Units | Note |
|---|---|---|---|---|---|
| Operating Temperature | $T_A$ | - 40 | 105 | °C | |
| Output Supply Voltage | ($V_{CC}$ - $V_{EE}$) | 15 | 30 | V | |
| Input Current (ON) | $I_{F(ON)}$ | 7 | 16 | mA | |
| Input Voltage (OFF) | $V_{F(OFF)}$ | - 3.6 | 0.8 | V | |

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**Table 5. Electrical Specifications (DC)**

All typical values are at $T_A$ = 25 °C, $V_{CC}$ - $V_{EE}$ = 30 V, $V_{EE}$ = Ground. All minimum and maximum specifications are at recommended operating conditions ($T_A$ = -40 to 105 °C, $I_{F(ON)}$ = 7 to 16 mA, $V_{F(OFF)}$ = -3.6 to 0.8 V, $V_{EE}$ = Ground, $V_{CC}$ = 15 to 30 V), unless otherwise noted.

| Parameter | Symbol | Min. | Typ. | Max. | Units | Test Conditions | Fig. | Note |
|---|---|---|---|---|---|---|---|---|
| High Level Peak Output Current | $I_{OH}$ | -1.0 | -2.3 | | A | $V_O = V_{CC} - 4$ V | 2, 3 | 5 |
| | | -2.0 | | | A | $V_{CC} - V_O \leq 15$ V | | 6 |
| Low Level Peak Output Current | $I_{OL}$ | 1.0 | 3.0 | | A | $V_O = V_{EE} + 2.5$ V | 5, 6 | 5 |
| | | 2.0 | | | A | $V_O - V_{EE} \leq 15$ V | | 7 |
| High Output Transistor RDS(ON) | $R_{DS,OH}$ | | 1.1 | 3.0 | Ω | $I_{OH}$ = -2.0 A | 7 | 8 |
| Low Output Transistor RDS(ON) | $R_{DS,OL}$ | | 0.55 | 1.8 | Ω | $I_{OL}$ = 2.0 A | 8 | 8 |
| High Level Output Voltage | $V_{OH}$ | Vcc-0.3 | Vcc – 0.2 | | V | $I_O$ = -100 mA | 1 | 9, 10 |
| High Level Output Voltage | $V_{OH}$ | | Vcc | | V | $I_O$ = 0 mA , $I_F$ = 10 mA | 1 | |
| Low Level Output Voltage | $V_{OL}$ | | 0.1 | 0.2 | V | $I_O$ = 100 mA | 4 | |
| High Level Supply Current | $I_{CCH}$ | | 1.9 | 5.0 | mA | Rg = 10 Ω, Cg = 25 nF, $I_F$ = 10 mA | 9, 10 | |
| Low Level Supply Current | $I_{CCL}$ | | 1.9 | 5.0 | mA | Rg = 10 Ω, Cg = 25 nF, $V_F$ = 0V | | |
| Threshold Input Current Low to High | $I_{FLH}$ | | 1.85 | 5.5 | mA | Rg = 10 Ω, Cg = 25 nF, $V_O$ > 5 V | 11, 12 | |
| Threshold Input Voltage High to Low | $V_{FHL}$ | 0.8 | | | V | | | |
| Input Forward Voltage | $V_F$ | 1.2 | 1.55 | 1.95 | V | $I_F$ = 10 mA | 17 | |
| Temperature Coefficient of Input Forward Voltage | $\Delta V_F/\Delta T_A$ | | -1.7 | | mV/°C | $I_F$ = 10 mA | | |
| Input Reverse Breakdown Voltage | $BV_R$ | 5 | | | V | $I_R$ = 100 µA | | |
| Input Capacitance | $C_{IN}$ | | 70 | | pF | f = 1 MHz, $V_F$ = 0 V | | |
| UVLO Threshold | $V_{UVLO+}$ | 12.1 | 12.8 | 13.5 | V | $V_O$ > 5 V, $I_F$ = 10 mA | | |
| | $V_{UVLO-}$ | 11.1 | 11.8 | 12.5 | | | | |
| UVLO Hysteresis | $UVLO_{HYS}$ | | 1.0 | | V | | | |

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### Table 6. Switching Specifications (AC)

All typical values are at $T_A$ = 25 °C, $V_{CC}$ - $V_{EE}$ = 30 V, $V_{EE}$ = Ground. All minimum and maximum specifications are at recommended operating conditions ($T_A$ = -40 to 105 °C, $I_{F(ON)}$ = 7 to 16 mA, $V_{F(OFF)}$ = -3.6 to 0.8 V, $V_{EE}$ = Ground, $V_{CC}$ = 15 to 30 V), unless otherwise noted.

| Parameter | Symbol | Min. | Typ. | Max. | Units | Test Conditions | Fig. | Note |
|---|---|---|---|---|---|---|---|---|
| Propagation Delay Time to High Output Level | $t_{PLH}$ | 50 | 110 | 300 | ns | Rg = 10 Ω, Cg = 25 nF, f = 20 kHz, Duty Cycle = 50%, $I_F$ = 7 mA to 16 mA, $V_{CC}$ = 15 V to 30 V | 13, 14, 15, 16, 18 | |
| Propagation Delay Time to Low Output Level | $t_{PHL}$ | 50 | 95 | 300 | ns | | | |
| Pulse Width Distortion | PWD | | 22 | 150 | ns | | | 11 |
| Propagation Delay Difference Between Any Two Parts | PDD ($t_{PHL}$ - $t_{PLH}$) | -200 | | 200 | ns | | 21, 22 | 12 |
| Propagation Delay Skew | $t_{PSK}$ | | | 150 | ns | | | 13 |
| Rise Time | $t_R$ | | 43 | | ns | $V_{CC}$ = 30 V | 18 | |
| Fall Time | $t_F$ | | 40 | | ns | | | |
| Output High Level Common Mode Transient Immunity | $\|CM_H\|$ | 25 | 35 | | kV/µs | $T_A$ = 25 °C, $I_F$ = 10 mA, $V_{CM}$ = 1500 V, $V_{CC}$ = 30 V, $V_{CM}$ = 1500 V | 19 | 14, 15 |
| Output Low Level Common Mode Transient Immunity | $\|CM_L\|$ | 25 | 35 | | kV/µs | $T_A$ = 25°C, $V_F$ = 0 V, $V_{CM}$ = 1500 V, $V_{CC}$ = 30 V, $V_{CM}$ = 1500 V | | 14, 16 |

### Table 7. Package Characteristics

All typical values are at $T_A$ = 25 °C. All minimum/maximum specifications are at recommended operating conditions, unless otherwise noted.

| Parameter | Symbol | Min. | Typ. | Max. | Units | Test Conditions | Fig. | Note |
|---|---|---|---|---|---|---|---|---|
| Input-Output Momentary Withstand Voltage* | $V_{ISO}$ | 3750 | | | $V_{RMS}$ | RH < 50%, t = 1 min., $T_A$ = 25 °C | | 17,18 |
| Input-Output Resistance | $R_{I-O}$ | | $10^{12}$ | | Ω | $V_{I-O}$ = 500 $V_{DC}$ | | 18 |
| Input-Output Capacitance | $C_{I-O}$ | | 0.6 | | pF | f =1 MHz | | |
| LED-to-Case Thermal Resistance | $\theta_{LC}$ | | 467 | | °C/W | | 23 | 19 |
| LED-to-Detector Thermal Resistance | $\theta_{LD}$ | | 442 | | | | | |
| Detector-to-Case Thermal Resistance | $\theta_{DC}$ | | 126 | | | | | |

* The Input-Output Momentary Withstand Voltage is a dielectric voltage rating that should not be interpreted as an input-output continuous voltage rating. For the continuous voltage rating, refer to your equipment level safety specification or Avago Technologies Application Note 1074 entitled "Optocoupler Input-Output Endurance Voltage."

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Notes:

1. Derate linearly above 70 °C free-air temperature at a rate of 0.3 mA/ °C.
2. Maximum pulse width = 10 µs. This value is intended to allow for component tolerances for designs with $I_O$ peak minimum = 2.0 A. See applications section for additional details on limiting IOH peak.
3. Derate linearly above 70 °C free-air temperature at a rate of 4.8 mW/ °C.
4. Derate linearly above 70 °C free-air temperature at a rate of 5.4 mW/ °C . The maximum LED junction temperature should not exceed 125 °C.
5. Maximum pulse width = 50 µs.
6. Output is sourced at -2.0 A with a maximum pulse width = 10 µs. $V_{CC}$-$V_O$ is measured to ensure 15 V or below.
7. Output is sourced at 2.0 A with a maximum pulse width = 10 µs. $V_O$-$V_{EE}$ is measured to ensure 15 V or below.
8. Output is sourced at -2.0 A/2.0 A with a maximum pulse width = 10 µs.
9. In this test $V_{OH}$ is measured with a dc load current. When driving capacitive loads, $V_{OH}$ will approach $V_{CC}$ as $I_{OH}$ approaches zero amps.
10. Maximum pulse width = 1 ms.
11. Pulse Width Distortion (PWD) is defined as $|t_{PHL}-t_{PLH}|$ for any given device.
12. The difference between $t_{PHL}$ and $t_{PLH}$ between any two QCPL-341H parts under the same test condition.
13. $t_{PSK}$ is equal to the worst-case difference in $t_{PHL}$ or $t_{PLH}$ that will be seen between units under the same test condition.
14. Pin 1 and 4 need to be connected to LED common. Split resistor network in the ratio 1.5:1 with 205 Ω at the anode and 137 Ω at the cathode.
15. Common mode transient immunity in the high state is the maximum tolerable $dV_{CM}/dt$ of the common mode pulse, $V_{CM}$, to assure that the output will remain in the high state (i.e., $V_O > 15.0$ V).
16. Common mode transient immunity in a low state is the maximum tolerable $dV_{CM}/dt$ of the common mode pulse, $V_{CM}$, to assure that the output will remain in a low state (i.e., $V_O < 1.0$ V).
17. In accordance with UL1577, each optocoupler is proof tested by applying an insulation test voltage ≤4500 $V_{RMS}$ for 1 second (leakage detection current limit, $I_{I-O} \leq 5$ µA).
18. Device considered a two-terminal device: pins 1, 2, 3 and 4 shorted together and pins 5, 6, 7 and 8 shorted together.
19. The device was mounted on a high conductivity test board as per JEDEC 51-7.

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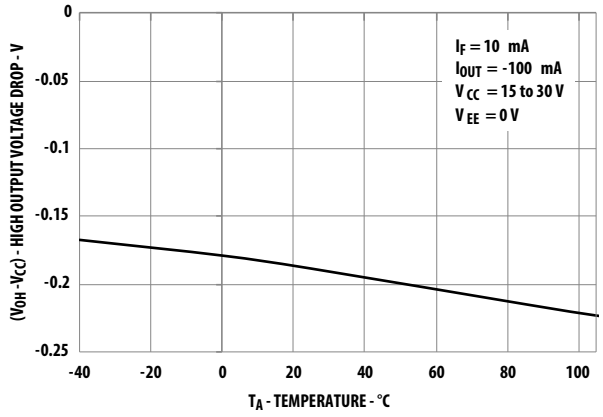



**Figure 1. $V_{OH}$ vs. temperature.**

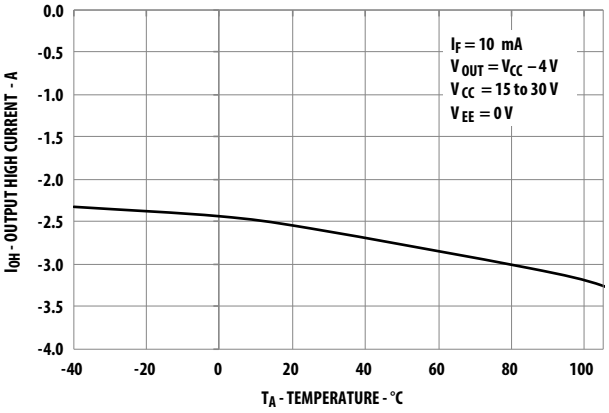



**Figure 2. $I_{OH}$ vs. temperature.**

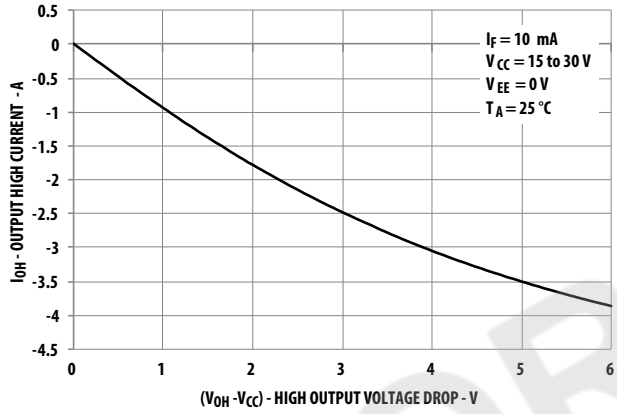



**Figure 3. $I_{OH}$ vs. $V_{OH}$.**

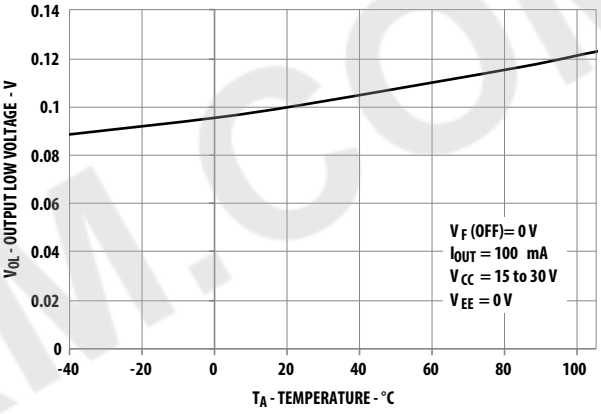



**Figure 4. $V_{OL}$ vs. Temperature.**

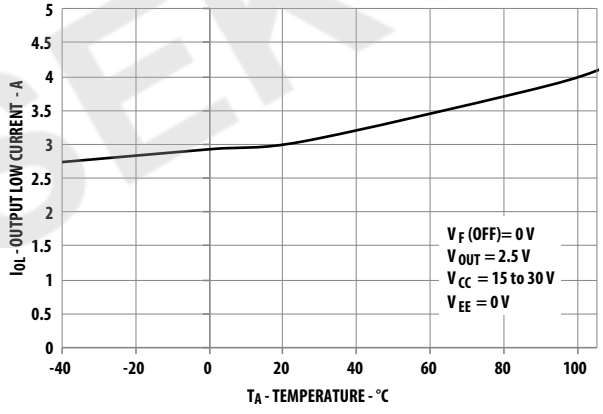



**Figure 5. $I_{OL}$ vs. temperature.**

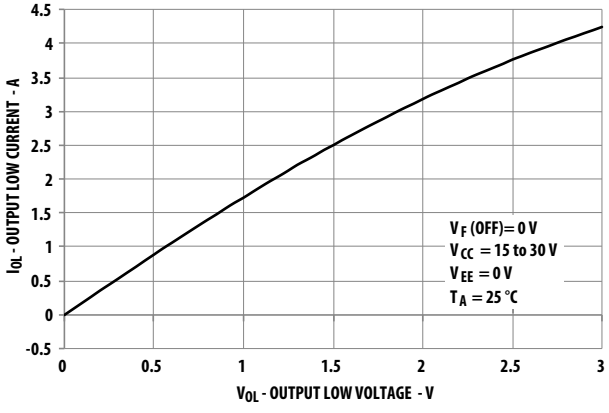



**Figure 6. $I_{OL}$ vs. $V_{OL}$.**

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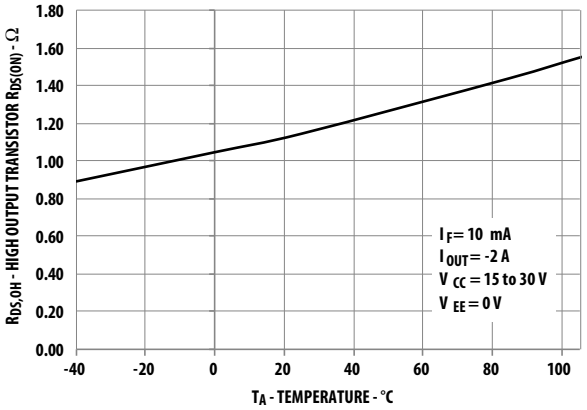



Figure 7. $R_{DS,OH}$ vs. temperature.

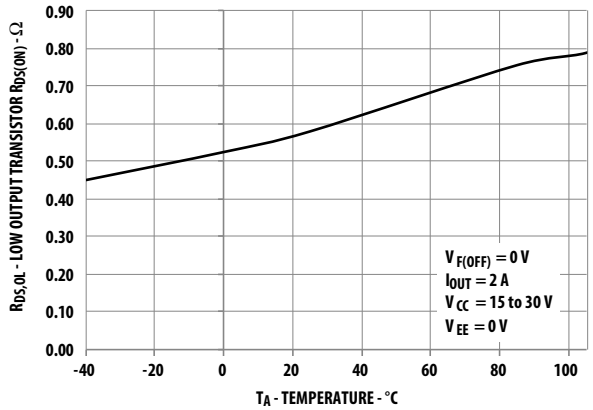



Figure 8. $R_{DS,OL}$ vs. temperature.

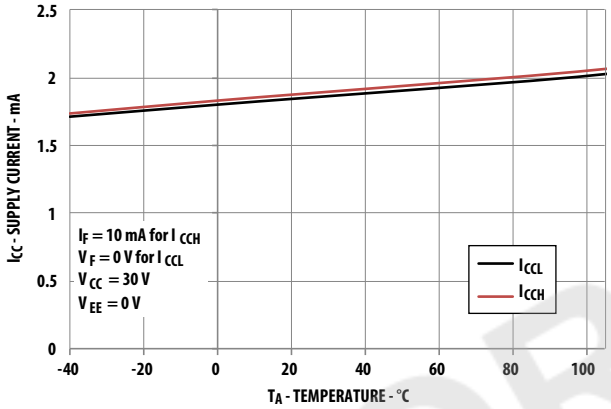



Figure 9. $I_{CC}$ vs. temperature.

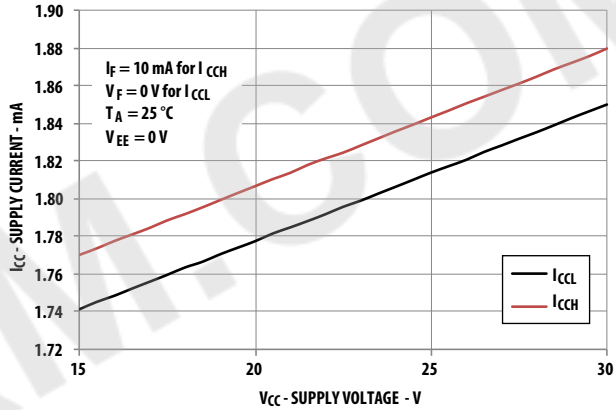



Figure 10. $I_{CC}$ vs. $V_{CC}$.

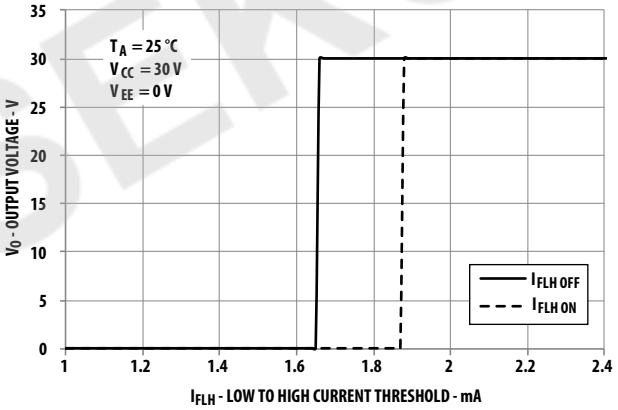



Figure 11. $I_{FLH}$ hysteresis.

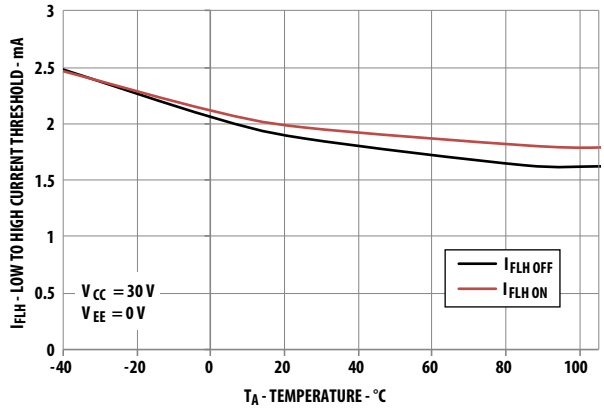



Figure 12. $I_{FLH}$ vs. temperature.

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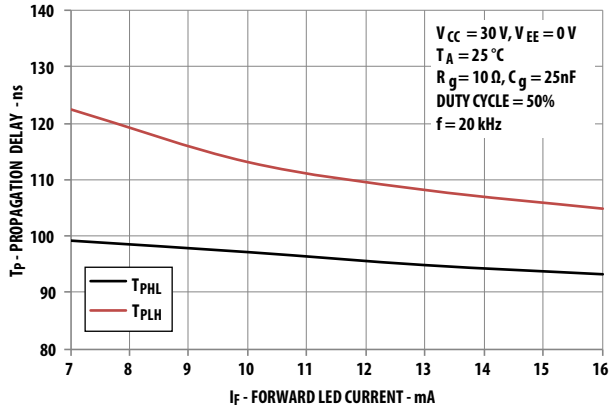



Figure 13. Propagation delay vs. $I_F$.

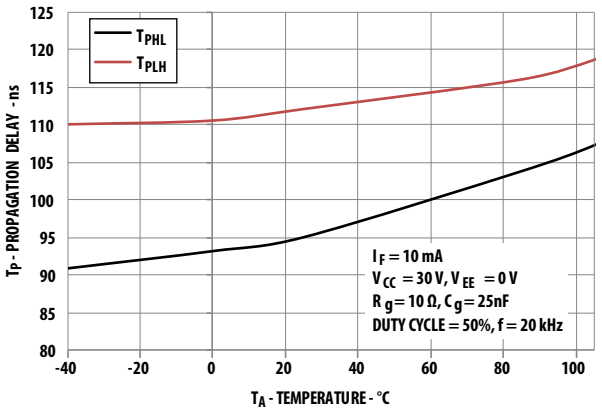



Figure 14. Propagation delay vs. temperature.

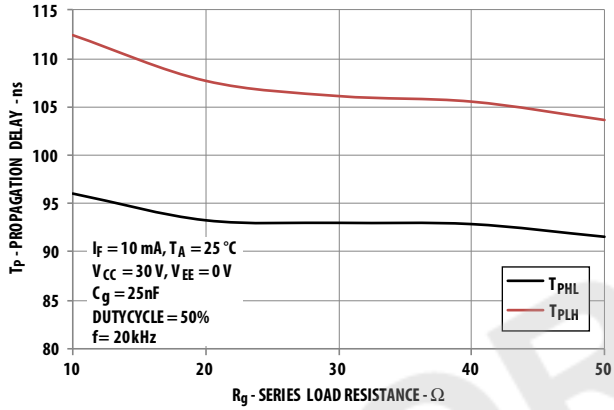



Figure 15. Propagation delay vs. Rg.

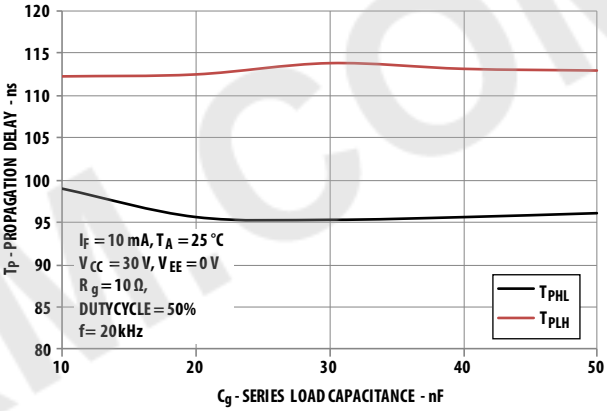



Figure 16. Propagation delay vs. Cg.

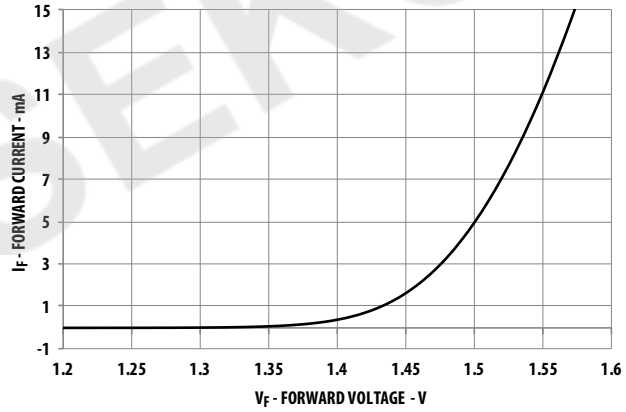
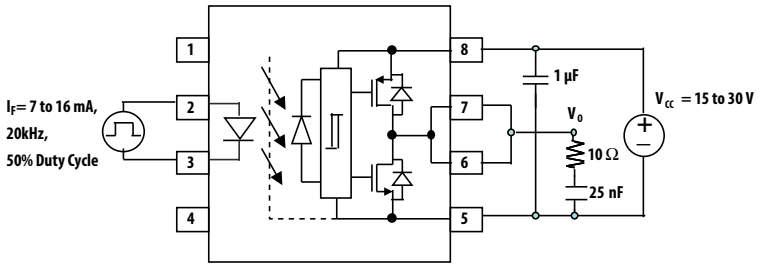



Figure 17. Input Current vs. forward voltage.

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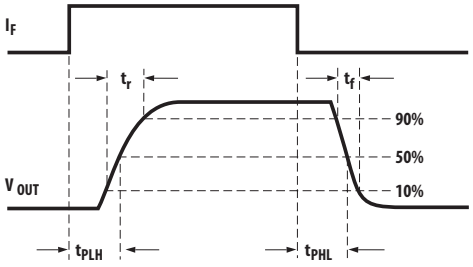
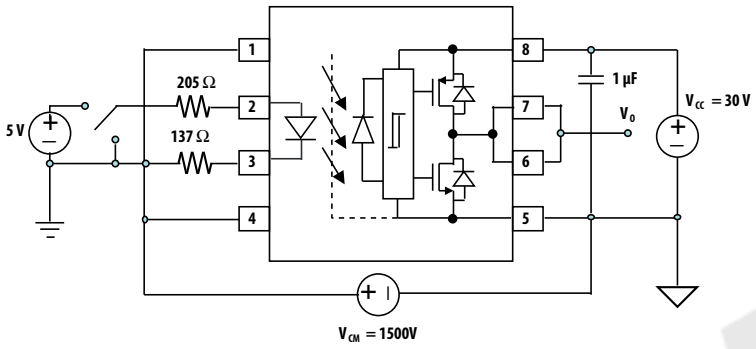




**Figure 18. $t_{PLH}$, $t_{PHL}$, tr and tf test circuit and waveforms.**



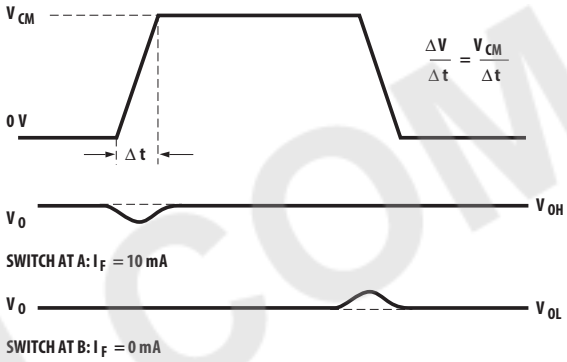




**Figure 19. CMR test circuit with split resistors network and waveforms.**

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## Application Information

### Recommended Application Circuit

The recommended application circuit shown in Figure 20 illustrates a typical gate drive implementation using the QCPL-341H. The following describes about driving IGBT. However, it is also applicable to MOSFET. Designers will need to adjust the $V_{CC}$ supply voltage, depending on the MOSFET or IGBT gate threshold requirements (Recommended $V_{CC}$ = 15 V for IGBT and 12 V for MOSFET).

The supply bypass capacitors (1 µF) provide the large transient currents necessary during a switching transition. Because of the transient nature of the charging currents, a low current (5.0 mA) power supply will be enough to power the device. The gate resistor $R_G$ serves to limit gate charge current and controls the IGBT collector voltage rise and fall times.

In PC board design, care should be taken to avoid routing the IGBT collector or emitter traces close to the QCPL-341H's inputs as this can result in unwanted coupling of transient signals into QCPL-341H and degrade performance.

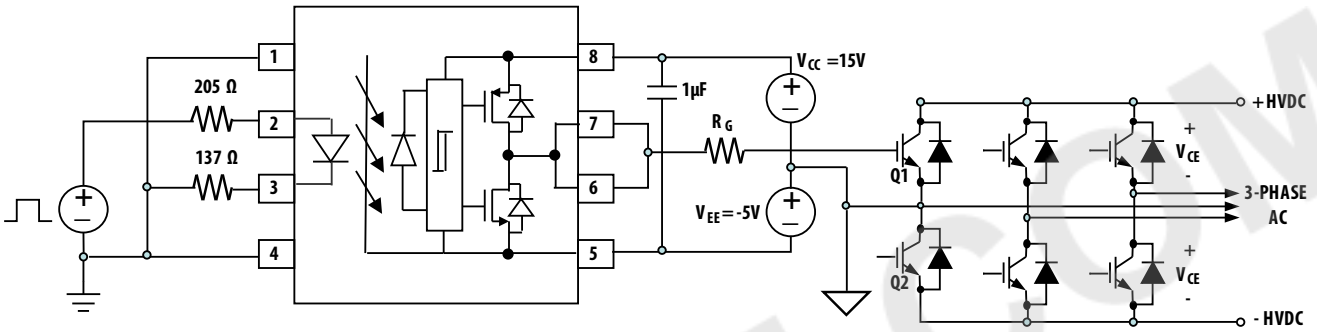



**Figure 20. Recommended application circuit with split resistors LED drive.**

### Selecting the Gate Resistor (Rg)

Step 1: Calculate Rg minimum from the $I_{OL}$ peak specification.
The IGBT and Rg in Figure 20 can be analyzed as a simple RC circuit with a voltage supplied by QCPL-341H.

$$Rg \geq \frac{V_{CC}-V_{EE}}{I_{OLPEAK}}$$
$$= \frac{15V - 5V}{2.5A}$$
$$= 8\Omega$$

Step 1: Check the QCPL-341H power dissipation and increase Rg if necessary.
The QCPL-341H total power dissipation (PT) is equal to the sum of the emitter power (PE) and the output power (PO).

$P_T = P_E + P_O$
$P_E = I_F \cdot V_F \cdot$ Duty Cycle
$P_O = P_{O(BIAS)} + P_{O(SWITCHING)}$
$= I_{CC} \cdot (V_{CC}-V_{EE}) + P_{HS} + P_{LS}$
$P_{HS} = (V_{CC}*Q_G*f) * R_{DS,OH(MAX)} / (R_{DS,OH(MAX)}+Rg) / 2$
$P_{LS} = (V_{CC}*Q_G*f) * R_{DS,OL(MAX)} / (R_{DS,OL(MAX)}+Rg) / 2$

Using $I_F$(worst case) = 16 mA, Rg = 8 Ω, Max Duty Cycle = 80%, $Q_G$ = 750 nC(1200V 100A IGBT),
f = 20 kHz and $T_A$ max = 70 °C:

$P_E = 16mA \cdot 1.95V \cdot 0.8 = 25mW$

$P_{HS} = (20V \cdot 750nC \cdot 20\ kHz) \cdot 3\Omega/(3\Omega+8\Omega)/2 = 40.91mW$
$P_{LS} = (20V \cdot 750nC \cdot 20\ kHz) \cdot 1.8\Omega/(1.8\Omega+8\Omega)/2 = 27.55mW$
$P_O = 5mA \cdot 20V + 40.9mW + 27.55mW$
$= 168.5mW < 250\ mW\ (P_{O(MAX)} @ 70\ °C)$

The value of 5 mA for $I_{CC}$ in the previous equation is the maximum $I_{CC}$ over the entire operating temperature range.

Since $P_O$ is less than $P_{O(MAX)}$, Rg = 8 Ω is alright for the power dissipation.

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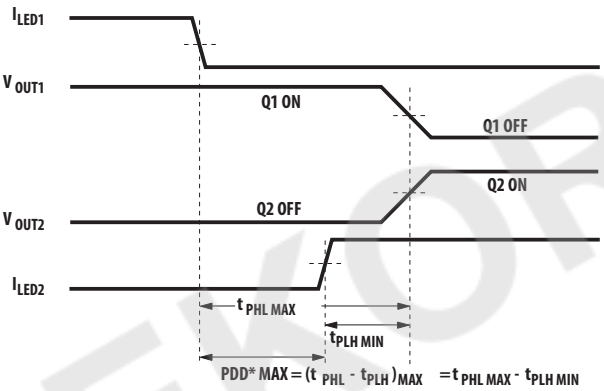
### Dead Time and Propagation Delay Specifications

The QCPL-341H includes a Propagation Delay Difference (PDD) specification intended to help designers minimize "dead time" in their power inverter designs. Dead time is the time period during which both the high and low side power transistors (Q1 and Q2 in Figure 20) are off. Any overlap in Q1 and Q2 conduction will result in large currents flowing through the power devices between the high and low voltage motor rails.

To minimize dead time in a given design, the turn on of LED2 should be delayed (relative to the turn off of LED1) so that under worst-case conditions, transistor Q1 has just turned off when transistor Q2 turns on, as shown in Figure 21. The amount of delay necessary to achieve this condition is equal to the maximum value of the propagation delay difference specification, $PDD_{MAX}$, which is specified to be 100 ns over the operating temperature range of 40 °C to 105 °C.

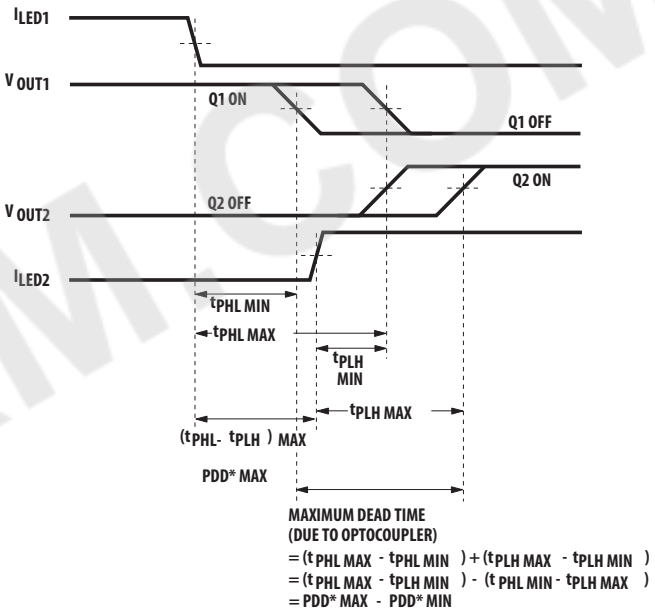
Delaying the LED signal by the maximum propagation delay difference ensures that the minimum dead time is zero, but it does not tell a designer what the maximum dead time will be. The maximum dead time is equivalent to the difference between the maximum and minimum propagation delay difference specifications as shown in Figure 22. The maximum dead time for the QCPL-341H is 400 ns (= 200 ns - (-200 ns)) over an operating temperature range of -40 °C to 105 °C.

Note that the propagation delays used to calculate PDD and dead time are taken at equal temperatures and test conditions since the optocouplers under consideration are typically mounted in close proximity to each other and are switching identical IGBTs.




*PDD = PROPAGATION DELAY DIFFERENCE
NOTE: FOR PDD CALCULATIONS THE PROPAGATION DELAYS ARE TAKEN AT THE SAME TEMPERATURE AND TEST CONDITIONS.

**Figure 21. Minimum LED skew for zero dead time.**




*PDD = PROPAGATION DELAY DIFFERENCE
NOTE: FOR DEAD TIME AND PDD CALCULATIONS ALL PROPAGATION DELAYS ARE TAKEN AT THE SAME TEMPERATURE AND TEST CONDITIONS.

**Figure 22. Waveforms for dead time.**

### LED Current Input with Hysteresis

The detector has optical receiver input stage with built in Schmitt trigger to provide logic compatible waveforms, eliminating the need for additional wave shaping. The hysteresis (Figure 11) provides differential mode noise immunity and minimizes the potential for output signal chatter.

### Under Voltage Lockout

The QCPL-341H Under Voltage Lockout (UVLO) feature is designed to prevent the application of insufficient gate voltage to the IGBT by forcing the QCPL-341H output low during power-up. IGBTs typically require gate voltages of 15 V to achieve their rated $V_{CE(ON)}$ voltage. At gate voltages below 13 V typically, the $V_{CE(ON)}$ voltage increases dramatically, especially at higher currents. At very low gate voltages (below 10 V), the IGBT may operate in the linear region and quickly overheat. The UVLO function causes the output to be clamped whenever insufficient operating supply ($V_{CC}$) is applied. Once $V_{CC}$ exceeds $V_{UVLO+}$ (the positive-going UVLO threshold), the UVLO clamp is released to allow the device output to turn on in response to input signals.

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**Thermal Model for QCPL-341H**

Definitions:

$T_{JE}$: LED junction temperature

$T_{JD}$: Detector IC junction temperature

$T_C$: Case temperature measured at the center of the package bottom

$\theta_{LC}$ : LED-to-case thermal resistance

$\theta_{LD}$: LED-to-detector thermal resistance

$\theta_{DC}$: Detector-to-case thermal resistance

$\theta_{CA}$: Case-to-ambient thermal resistance

*$\theta_{CA}$ will depend on the board design and the placement of the part

$T_A$: Ambient temperature.

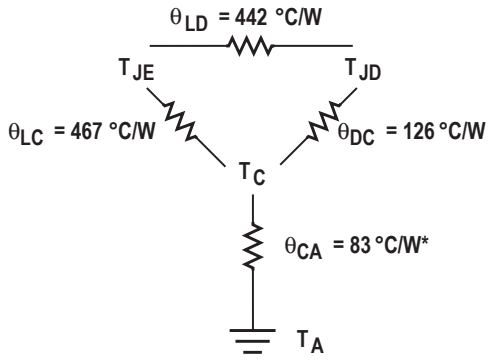



**Figure 23. Thermal model.**

The steady state thermal model for the QCPL-341H is shown in Figure 23. The thermal resistance values given in this model can be used to calculate the temperatures at each node for a given operating condition. As shown by the model, all heat generated flows through $\theta_{CA}$ which raises the case temperature $T_C$ accordingly. The value of $\theta_{CA}$ depends on the conditions of the board design and is, therefore, determined by the designer. The value of $\theta_{CA}$ = 83°C/W was obtained from thermal measurements using a 2.5 x 2.5 inch PC board, with small traces (no ground plane), a single QCPL-341H soldered into the center of the board and still air. The absolute maximum power dissipation derating specifications assume a $\theta_{CA}$ value of 83°C/W.

From the thermal mode in Figure 23 the LED and detector IC junction temperatures can be expressed as:

$$T_{JE} = P_E \bullet \left(\theta_{LC} \,||\, \left(\theta_{LD} + \theta_{DC}\right) + \theta_{CA}\right) + P_D \bullet \left(\frac{\theta_{LC} \bullet \theta_{DC}}{\theta_{LC} + \theta_{DC} + \theta_{LD}} + \theta_{CA}\right) + T_A$$

$$T_{JD} = P_E \bullet \left(\frac{\theta_{LC} \bullet \theta_{DC}}{\theta_{LC} + \theta_{DC} + \theta_{LD}} + \theta_{CA}\right) + P_D \bullet \left(\theta_{DC} \,||\, \left(\theta_{LD} + \theta_{LC}\right) + \theta_{CA}\right) + T_A$$

## Related Documents

AV02-0421EN – AN5336 – Gate Drive Optocoupler Basic Design for IGBT / MOSFET

AV02-3698EN – AN1043 – Common-Mode Noise: Sources and Solutions

AV02-0310EN – Reliability Data - Plastics Optocouplers Product ESD and Moisture Sensitivity

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